

Classification using Logistic Regression Artificial Intelligence & Machine Learning Internship - Task 4

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1. Introduction

Machine learning classification is widely used in healthcare to support diagnosis and decision making. In this project the Breast Cancer Wisconsin Dataset was used to classify tumors as benign or malignant. Logistic Regression was selected as the primary classifier because it is simple, interpretable and highly effective for binary classification. A Decision Tree Classifier was also trained for comparison, allowing evaluation of two different learning approaches. The complete workflow demonstrates preprocessing, visualization, model training, evaluation and interpretation, similar to real-world machine learning pipelines.

2. Objectives

The objectives were to understand the dataset, perform exploratory data analysis, preprocess features, apply feature scaling, build a Logistic Regression classifier, evaluate performance using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC Curve and AUC Score, compare results with a Decision Tree Classifier and identify the best-performing model.

3. Dataset Description

The Breast Cancer Wisconsin Dataset available in Scikit-Learn contains 569 observations with 30 numerical features extracted from digitized images of breast mass cell nuclei. The target contains two classes: Malignant (0) and Benign (1). The dataset contains no missing values and is considered an excellent benchmark dataset for binary classification.

4. Methodology

The workflow started by importing Python libraries including NumPy, Pandas, Matplotlib, Seaborn and Scikit-Learn. The dataset was loaded into a DataFrame and explored using `head()`, `info()` and `describe()`. Missing values were verified, target distribution was visualized and feature correlation was studied using a heatmap. Data was split into training and testing sets, followed by feature scaling with `StandardScaler`. Logistic Regression was trained, predictions were generated and evaluation metrics were computed. Finally, a Decision Tree model was trained using the same split and compared against Logistic Regression.

5. Data Preprocessing

Feature scaling is essential for Logistic Regression because features with different ranges may affect optimization. `StandardScaler` transformed each feature to zero mean and unit variance, improving convergence and ensuring balanced feature influence. Train-test splitting prevented information leakage and enabled fair evaluation on unseen data.

6. Model Evaluation

Accuracy measured the proportion of correct predictions. Precision quantified how many predicted positives were actually positive. Recall measured the ability to identify positive cases. F1-Score balanced Precision and Recall. The Confusion Matrix visualized correct and incorrect predictions. ROC Curve and AUC Score evaluated discrimination capability across thresholds, while the Precision-Recall Curve provided additional insight into classifier performance.

7. Visual Analysis

The notebook includes Dataset Preview, Target Distribution, Correlation Heatmap, Confusion Matrix, ROC Curve, Precision-Recall Curve, Feature Importance, Prediction Distribution, Accuracy Comparison, ROC Comparison and Model Comparison charts. Insert screenshots of each visualization in the final report to demonstrate project execution and improve presentation quality.

8. Results

Both models successfully classified breast cancer samples. Logistic Regression achieved strong performance after feature scaling and served as an excellent baseline classifier. The Decision Tree classifier was also evaluated, allowing comparison of predictive capability and interpretability. Include the final Accuracy, Precision, Recall, F1-Score and AUC values from your notebook in the table below.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	_____	_____	_____	_____	_____
Decision Tree	_____	_____	_____	_____	_____

9. Discussion

The comparison of two classifiers demonstrates that selecting an appropriate model requires more than measuring accuracy alone. Metrics such as Precision, Recall, F1-Score and AUC provide a more complete understanding of model behaviour. Visualizations further explain strengths and weaknesses, making the workflow transparent and reproducible.

10. Conclusion

This project successfully demonstrated an end-to-end machine learning classification workflow. From data exploration and preprocessing to model comparison and evaluation, every stage reflected industry practices. Logistic Regression proved to be a reliable classifier for this dataset, while Decision Tree offered an alternative perspective. The project strengthened practical skills in supervised learning, evaluation metrics, feature scaling and visualization.

11. Future Work

Future improvements include Hyperparameter Tuning, Cross Validation, Random Forest, Support Vector Machine, XGBoost, PCA, Feature Selection and deployment using Streamlit or Flask. These enhancements can improve predictive performance and convert the notebook into a deployable machine learning application.